

COURSE NOTES

---

# Lie Theory

---

**Instructor:**

Prof. Efilm ZELMANOV

*Student:*

Leo

March 7, 2026

# Contents

|          |                                                                 |           |
|----------|-----------------------------------------------------------------|-----------|
| <b>1</b> | <b>Lecture 1 Where Lie algebras come from</b>                   | <b>2</b>  |
| 1.1      | Introduction . . . . .                                          | 2         |
| 1.2      | Where do Lie algebras come from? . . . . .                      | 5         |
| 1.2.1    | They come from associative algebras . . . . .                   | 5         |
| 1.2.2    | They come from involutions of central simple algebras . . . . . | 6         |
| 1.2.3    | They come from the derivation of any algebra . . . . .          | 12        |
| 1.2.4    | They come from brackets . . . . .                               | 18        |
| 1.2.5    | They come from Lie groups . . . . .                             | 20        |
| 1.3      | Basic concepts of Lie algebra . . . . .                         | 22        |
| <b>2</b> | <b>Lecture 2 Engel Theorem</b>                                  | <b>25</b> |
| 2.1      | Engel Theorem the first type . . . . .                          | 25        |
| 2.2      | Engel Theorem the second type . . . . .                         | 26        |

# Chapter 1

## Lecture 1 Where Lie algebras come from

### 1.1 Introduction

Lie algebras were originally introduced by Sophus Lie in the 1870s to study the concept of continuous transformation groups. Lie's motivation was to develop a theory for differential equations analogous to Galois theory for algebraic equations. He discovered that the infinitesimal transformations of a continuous group form a linear space equipped with a bracket operation, now known as a Lie algebra. This linearization allows complex global geometric problems to be translated into algebraic problems, making them more tractable.

Let  $F$  be a field (mostly  $\text{char } F = 0$ , even  $F = \mathbb{C}$ ).

**Definition 1.1.1.** A linear algebra  $A$  is a vector space over  $F$ , with a binary bilinear operation:

- *binary operation:*  $A \times A \rightarrow A$ ,  $(a, b) \mapsto ab$ .
- *bilinear:* this operation is linear in each variable. In other words,  $\forall \alpha \in F$  and  $a, a', a'', b, b', b'' \in A$ , we have:

$$(\alpha a)b = a(\alpha b) = \alpha(ab)$$

$$(a' + a'')b = a'b + a''b$$

$$a(b' + b'') = ab' + ab''$$

The algebra  $A$  is associative if  $\forall a, b, c \in A$ ,  $(ab)c = a(bc)$ .

**Example 1.1.2.**  $M_n(F)$ , the algebra of  $n \times n$  matrices over  $F$ .

**Example 1.1.3.** Let  $V$  be a vector space,  $\text{End}_F(V) = \{\text{linear transformations } \varphi : V \rightarrow V\}$  is an associative algebra with the composition operation  $(\phi\psi)(v) = \phi(\psi(v))$ .

If  $\dim V < \infty$ , when we fix a basis, then we get a matrix representation of  $\varphi$  in our fix basis.

The most important thing everyone should know is Wedderburn's theorem.

**Theorem 1.1.4** (Wedderburn-Artin). *Let  $A$  be a semisimple Artinian ring (with 1). Then  $A$  is a direct sum of matrix rings over division rings. Specifically, if  ${}_A A \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$  where  $S_1, \dots, S_r$  are non-isomorphic simple modules occurring with multiplicities  $n_1, \dots, n_r$ , then*

$$A \simeq M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

where  $D_i = \text{End}_A(S_i)^{op}$ . This is a ring isomorphism, and each  $M_{n_i}(D_i)$  is a 2-sided ideal.

*Proof.* First, since  $A$  is a semisimple ring, its left regular module  ${}_A A$  can be decomposed into a direct sum of simple left  $A$ -modules:

$${}_A A \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$$

where  $S_1, \dots, S_r$  are pairwise non-isomorphic simple modules.

By Schur's lemma, the endomorphism ring of a simple module is a division ring, so  $D_i = \text{End}_A(S_i)^{op}$  is a division ring.

Next, we compute the endomorphism ring of the left regular module  $\text{End}_A({}_A A)$ . On one hand, for any ring  $A$  with 1, we have the ring isomorphism  $\text{End}_A({}_A A) \simeq A^{op}$ . On the other hand, using our direct sum decomposition:

$$\text{End}_A({}_A A) \simeq \text{End}_A(S_1^{n_1}) \oplus \cdots \oplus \text{End}_A(S_r^{n_r})$$

This decomposition holds because  $\text{Hom}_A(S_i^{n_i}, S_j^{n_j}) = 0$  whenever  $i \neq j$ .

For each component in the direct sum, we can rewrite the endomorphism ring using matrices over  $D_i$ :

$$\text{End}_A(S_i^{n_i}) \simeq M_{n_i}(D_i)^{op} \simeq M_{n_i}(D_i^{op})$$

Substituting this back into our equation for  $A^{op}$ , we obtain:

$$A^{op} \simeq M_{n_1}(D_1^{op}) \oplus \cdots \oplus M_{n_r}(D_r^{op})$$

To recover the ring  $A$ , we take the opposite ring on both sides. Recall the property that  $M_n(R)^{op} \simeq M_n(R^{op})$ . Applying this to each block yields:

$$A \simeq (A^{op})^{op} \simeq \left( \bigoplus_{i=1}^r M_{n_i}(D_i^{op}) \right)^{op} \simeq \bigoplus_{i=1}^r M_{n_i}(D_i)$$

Thus, the isomorphism is established.  $\square$

**Remark 1.1.5.** If the base field  $F$  is algebraically closed, every finite-dimensional division algebra over  $F$  is equal to  $F$ . Hence the theorem simplifies to

$$A \cong \prod_{i=1}^r M_{n_i}(F).$$

**Example 1.1.6.** Let  $G$  be a finite group and let  $\mathbb{C}[G]$  be its group algebra. By Maschke's theorem  $\mathbb{C}[G]$  is semisimple. Therefore

$$\mathbb{C}[G] \cong \prod_{i=1}^r M_{n_i}(\mathbb{C}),$$

where  $n_i$  are the dimensions of the irreducible complex representations of  $G$ .

**Definition 1.1.7.** Suppose  $F$  is algebraically closed. An algebra  $A$  is simple if

1.  $\nexists I \triangleleft A$  and  $I \neq 0$ .
2.  $A \cdot A \neq (0)$ .

**Corollary 1.1.8.** Any simple finite dimensional associative algebra is isomorphic to  $M_n(F)$  for some  $n$ .

**Definition 1.1.9** (Jacobson Radical). Let  $A$  be an associative algebra with identity over a field  $F$ . The Jacobson radical of  $A$ , denoted by  $\text{Rad}(A)$  or  $J(A)$ , is defined as the intersection of all maximal left ideals of  $A$ :

$$\text{Rad}(A) = \bigcap_{M \text{ maximal left ideal of } A} M.$$

**Proposition 1.1.10.** Let  $A$  be an associative algebra with identity. Then

$$\text{Rad}(A) = \bigcap_{S \text{ simple left } A\text{-module}} \text{Ann}_A(S),$$

where

$$\text{Ann}_A(S) = \{a \in A \mid aS = 0\}$$

is the annihilator of  $S$  in  $A$ .

**Proposition 1.1.11** (Equivalent Characterization). Let  $A$  be an associative algebra with identity. Then an element  $a \in A$  lies in  $\text{Rad}(A)$  if and only if for every  $x \in A$  the element

$$1 - ax$$

is invertible in  $A$ .

**Proposition 1.1.12.** *The Jacobson radical  $\text{Rad}(A)$  is a two-sided ideal of  $A$ .*

**Definition 1.1.13** (Semisimple Algebra). *An associative algebra  $A$  is called semisimple if*

$$\text{Rad}(A) = 0.$$

## 1.2 Where do Lie algebras come from?

**Definition 1.2.1.** *A Lie algebra is a vector space  $L$  over a field  $F$ , with a binary operation  $[a, b] \in L$ , such that*

1.  $[a, b]$  is bilinear.
2.  $[a, b] = -[b, a]$ .
3.  $[a, [b, c]] + [c, [a, b]] + [b, [c, a]] = 0$ , this is called Jacobi identity.

### 1.2.1 They come from associative algebras

Let  $A$  be an associative algebra.  $\forall a, b \in A$ , define  $[a, b] = ab - ba$ . Check that  $(A, [ , ])$  is a Lie algebra:

*Proof.* Bilinearity and skew-symmetry are clear. For Jacobi identity:

$$\begin{aligned} & [a, [b, c]] + [b, [c, a]] + [c, [a, b]] \\ &= a(bc - cb) - (bc - cb)a + b(ca - ac) - (ca - ac)b + c(ab - ba) - (ab - ba)c \\ &= abc - acb - bca + cba + bca - bac - cab + acb + cab - cba - abc + bac \\ &= 0. \end{aligned}$$

□

We denote this new algebra  $(A, [ , ])$  by  $A^{(-)}$ .

Suppose that  $L \subseteq A$  and  $[L, L] \subseteq L$ . Then  $L \leq A^{(-)}$ .

**Example 1.2.2.**  $M_n(F)^{(-)} = \mathfrak{gl}(n)$ . We call it the general linear Lie algebra.  $[L, L] = \{\sum_i [a_i, b_i]\} \subseteq L$ . Clearly  $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq \mathfrak{gl}(n)$  and for  $n \geq 2$ ,  $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq 0$ .

**Example 1.2.3.** Focus on  $[\mathfrak{gl}(n), \mathfrak{gl}(n)]$ , all matrices are of trace 0. This is called the special linear Lie algebra, denoted by  $\mathfrak{sl}(n)$ .

**Example 1.2.4.** *If the base field changed, some properties of the Lie algebra may change. For example, if  $\text{char } F = 2$ , then  $[a, b] = [b, a]$ . The Lie bracket is no longer skew-symmetric but actually symmetric.*

*Consider  $[\mathfrak{gl}(n), \mathfrak{gl}(n)] = \mathfrak{sl}(n)$ . If  $\text{char } F = 0$  or  $\text{char } F \nmid n$ , then  $\mathfrak{sl}(n)$  is a simple Lie algebra. If  $\text{char } F \mid n$ , then  $\mathfrak{sl}(n)$  is not simple.*

**Definition 1.2.5.**  $Z$  is the center of Lie algebra  $L$  if  $Z = \{z \in L \mid [z, L] = 0\}$ .

### 1.2.2 They come from involutions of central simple algebras

We can also find Lie algebras from the involution of a central simple algebra. Let  $A$  be central simple algebra and  $*$  :  $A \rightarrow A$  be a linear transformation.  $*$  is called a an involution if  $(a^*)^* = a$  and  $(ab)^* = b^*a^*$ ,  $\forall a, b \in A$ .

**Proposition 1.2.6.** *Let  $A$  be a central simple algebra over a field  $F$  equipped with an involution  $*$  :  $A \rightarrow A$ . Then the set*

$$L = \{x \in A \mid *(x) = -x\}$$

*forms a Lie algebra under the commutator bracket*

$$[x, y] = xy - yx.$$

*Proof.* First observe that any associative algebra  $A$  becomes a Lie algebra with the commutator bracket

$$[x, y] = xy - yx.$$

Let

$$L = \{x \in A \mid *(x) = -x\}.$$

We show that  $L$  is closed under the Lie bracket.

Take  $x, y \in L$ . Then

$$*(x) = -x, \quad *(y) = -y.$$

Since  $*$  is an involution, it satisfies

$$*(xy) = *(y) * (x).$$

Hence

$$*([x, y]) = *(xy - yx) = *(y) * (x) - *(x) * (y).$$

Substituting  $*(x) = -x$  and  $*(y) = -y$ , we obtain

$$*([x, y]) = (-y)(-x) - (-x)(-y) = yx - xy = -[x, y].$$

Thus  $[x, y] \in L$ . Therefore  $L$  is closed under the commutator bracket.

Since the commutator bracket in any associative algebra satisfies bilinearity, antisymmetry, and the Jacobi identity, it follows that  $L$  is a Lie algebra.  $\square$

This construction explains why many classical Lie algebras arise from central simple algebras with involution. In particular, the classical Lie algebras of Cartan–Killing type can be obtained in this way.

**Example 1.2.7** (Orthogonal involution). *Let*

$$A = M_n(F)$$

*and define the involution*

$$*(X) = X^T.$$

*Then*

$$L = \{X \in M_n(F) \mid X^T = -X\}.$$

*This is the Lie algebra of skew-symmetric matrices, which is the orthogonal Lie algebra*

$$\mathfrak{so}_n(F).$$

**Example 1.2.8** (Symplectic involution). *Let*

$$A = M_{2n}(F)$$

*and let*

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

*Define*

$$*(X) = J^{-1}X^TJ.$$

*Then*

$$L = \{X \in M_{2n}(F) \mid *(X) = -X\}$$

*is the symplectic Lie algebra*

$$\mathfrak{sp}_{2n}(F).$$

**Example 1.2.9** (Unitary involution). *Let  $K/F$  be a quadratic field extension with Galois involution  $a \mapsto \bar{a}$ . Consider*

$$A = M_n(K)$$

*with involution*

$$*(X) = \overline{X}^T.$$

Then

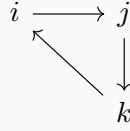
$$L = \{X \in M_n(K) \mid *(X) = -X\}$$

is the unitary Lie algebra

$$\mathfrak{su}_n.$$

Therefore classical Lie algebras arise from pairs  $(A, *)$  where  $A$  is a central simple algebra and  $*$  is an involution. This provides an algebraic framework for the Cartan–Killing classification of classical Lie algebras.

**Example 1.2.10.** The Quatanion algebra is an associative algebra with the involution  $a \mapsto a^*$ . It is 4 dimensional central simple algebra with standard basis  $1, i, j, k$ , where the relations are  $i^2 = j^2 = k^2 = -1, ij = k, jk = i, ki = j$  as the following graph:



We use  $\mathbb{H}$  to denote the Quatanion algebra when we choose  $\mathbb{R}$  as the base field. i.e.  $\mathbb{H} = \{a_0 1 + a_1 i + a_2 j + a_3 k \mid a_0, a_1, a_2, a_3 \in \mathbb{R}\}$ .  $\forall x \in \mathbb{H}$  where  $x = a_0 1 + a_1 i + a_2 j + a_3 k$ ,  $*$  is defined as  $x^* = a_0 1 - a_1 i - a_2 j - a_3 k$  and  $x^{-1} = \frac{1}{a_0^2 + a_1^2 + a_2^2 + a_3^2} (a_0 1 - a_1 i - a_2 j - a_3 k)$ . Hence  $\mathbb{H}$  is a division algebra.

If the base field is algebraically closed,  $F = \overline{F}$ . We can denote the Quatanion algebra by  $2 \times 2$  matrices

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix}$$

Where  $\alpha, \beta, \gamma, \xi \in F$ . In this case it is no longer a division algebra ( $a_0^2 + a_1^2 + a_2^2 + a_3^2$  may equal to 0), but just  $M_2(F)$ . And in this case the involution  $*$  is defined as:

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix} \mapsto \begin{pmatrix} \xi & -\beta \\ -\gamma & \alpha \end{pmatrix}$$

Which is the Symplectic involution.

**Example 1.2.11.** From what we discussd. For a  $2n \times 2n$  matrix  $A = \begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix}$  in which  $A_1, A_2, A_3, A_4$  are  $n \times n$  matrices, the unitary involution is defined as:

$$A^* = \begin{pmatrix} A_4^t & -A_3^t \\ -A_2^t & A_1^t \end{pmatrix}$$

And the unitary involution is defined as:

$$A^* = \overline{A}^t$$

When  $n = 1$  we have two different type, and we will get two different Lie algebras. But they are isomorphic.

**Proposition 1.2.12.** *Let  $F$  be a field of characteristic  $\neq 2$ , and let*

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in M_2(F).$$

*Define an involution  $*$  on  $M_2(F)$  by*

$$X^* = J^{-1}X^tJ.$$

*Then the Lie algebra of  $*$ -skew elements,*

$$K(M_2(F), *) = \{X \in M_2(F) \mid X^* = -X\},$$

*is exactly  $\mathfrak{sl}_2(F)$ .*

*Proof.* Let

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F).$$

Since

$$J^{-1} = -J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

we compute

$$X^* = J^{-1}X^tJ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Now the condition  $X^* = -X$  becomes

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix}.$$

Hence  $d = -a$ , while  $b$  and  $c$  are arbitrary. Therefore

$$K(M_2(F), *) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \mid a, b, c \in F \right\}.$$

But this is precisely the set of all  $2 \times 2$  matrices of trace 0, namely

$$\mathfrak{sl}_2(F) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \mid a, b, c \in F \right\}.$$

Thus

$$K(M_2(F), *) = \mathfrak{sl}_2(F).$$

Since  $K(M_2(F), *)$  is closed under the commutator bracket, it follows that the Lie algebra obtained from the symplectic involution on  $M_2(F)$  is  $\mathfrak{sl}_2(F)$ .  $\square$

**Example 1.2.13.** Let  $K/F$  be a quadratic field extension, and let  $a \mapsto \bar{a}$  denote the nontrivial  $F$ -automorphism of  $K$ . Consider the algebra  $M_2(K)$  and define an involution  $*$  by

$$X^* = \overline{X}^t.$$

Then  $*$  is a unitary involution on  $M_2(K)$ .

The Lie algebra of  $*$ -skew elements is

$$K(M_2(K), *) = \{X \in M_2(K) \mid X^* = -X\}.$$

If

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then

$$X^* = \overline{X}^t = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}.$$

Thus the condition  $X^* = -X$  is equivalent to

$$\bar{a} = -a, \quad \bar{d} = -d, \quad \bar{c} = -b.$$

Hence every element of  $K(M_2(K), *)$  has the form

$$X = \begin{pmatrix} a & b \\ -\bar{b} & d \end{pmatrix}, \quad \text{with } \bar{a} = -a, \bar{d} = -d.$$

This is the unitary Lie algebra associated with the standard hermitian form.

Its derived Lie algebra consists of those elements with trace 0, namely

$$\mathfrak{su}_2(K/F) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & -a \end{pmatrix} \mid a, b \in K, \bar{a} = -a \right\}.$$

After extending scalars to a splitting field (for instance to an algebraic closure of  $F$ ), this Lie algebra becomes isomorphic to  $\mathfrak{sl}_2$ . Therefore the Lie algebra arising from a unitary involution in the  $2 \times 2$  case is of type  $A_1$ .

**Remark 1.2.14.** Why does this Lie algebra become  $\mathfrak{sl}_2$ ? The reason is that unitary Lie algebras are forms of special linear Lie algebras. In the present  $2 \times 2$  case, the corresponding type is  $A_1$ . Over a field where the quadratic extension splits, the involution becomes equivalent to the ordinary transpose situation, and the resulting special unitary Lie algebra becomes the split simple Lie algebra of type  $A_1$ , namely

$$\mathfrak{sl}_2.$$

Thus, although over the ground field one naturally writes the Lie algebra as  $\mathfrak{su}_2$ , its split form is precisely  $\mathfrak{sl}_2$ .

**Example 1.2.15** (The unitary involution on  $M_2(\mathbb{C})$  and the Lie algebra  $\mathfrak{su}_2$ ). Let  $F = \mathbb{R}$  and  $K = \mathbb{C}$ , and let  $\bar{\phantom{x}}$  denote complex conjugation. Consider the algebra  $M_2(\mathbb{C})$  equipped with the involution

$$X^* = \overline{X}^t.$$

This is the standard unitary involution on  $M_2(\mathbb{C})$  relative to the extension  $\mathbb{C}/\mathbb{R}$ . We first determine the Lie algebra of  $*$ -skew elements:

$$K(M_2(\mathbb{C}), *) = \{X \in M_2(\mathbb{C}) \mid X^* = -X\}.$$

Let

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{C}).$$

Then

$$X^* = \overline{X}^t = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}.$$

Thus the condition  $X^* = -X$  is equivalent to

$$\bar{a} = -a, \quad \bar{d} = -d, \quad \bar{c} = -b.$$

Hence  $a$  and  $d$  are purely imaginary, and  $c = -\bar{b}$ . Therefore

$$K(M_2(\mathbb{C}), *) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & d \end{pmatrix} \mid a, d \in i\mathbb{R}, b \in \mathbb{C} \right\}.$$

This is the unitary Lie algebra  $\mathfrak{u}_2$ .

Its derived Lie algebra is the trace-zero part, namely

$$\mathfrak{su}_2 = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & -a \end{pmatrix} \mid a \in i\mathbb{R}, b \in \mathbb{C} \right\}.$$

Indeed,  $\text{tr}([X, Y]) = 0$  for all  $X, Y \in M_2(\mathbb{C})$ , so every commutator in  $\mathfrak{u}_2$  has trace 0. Conversely, in this  $2 \times 2$  case, the trace-zero subspace is generated by commutators, and hence it is exactly the derived Lie algebra.

To see that  $\mathfrak{su}_2$  is 3-dimensional over  $\mathbb{R}$ , write

$$a = it, \quad b = u + iv, \quad t, u, v \in \mathbb{R}.$$

Then every element of  $\mathfrak{su}_2$  has the form

$$\begin{pmatrix} it & u + iv \\ -u + iv & -it \end{pmatrix}, \quad t, u, v \in \mathbb{R}.$$

Therefore

$$\dim_{\mathbb{R}} \mathfrak{su}_2 = 3.$$

A convenient real basis is

$$H = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

One checks directly that

$$[H, X] = 2Y, \quad [H, Y] = -2X, \quad [X, Y] = 2H,$$

so  $\mathfrak{su}_2$  is a 3-dimensional real Lie algebra.

Now we explain why this Lie algebra becomes  $\mathfrak{sl}_2(\mathbb{C})$  after extending scalars from  $\mathbb{R}$  to  $\mathbb{C}$ . Consider the complexification

$$\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}.$$

Since  $H, X, Y$  form a real basis of  $\mathfrak{su}_2$ , they also form a complex basis of  $\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}$ .

Define

$$h = -iH = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad e = \frac{1}{2}(X - iY) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \frac{1}{2}(-X - iY) = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}.$$

These elements lie in  $\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}$  and satisfy

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

Thus they form a standard  $\mathfrak{sl}_2$ -triple, and therefore

$$\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{sl}_2(\mathbb{C}).$$

In other words,  $\mathfrak{su}_2$  is a real form of  $\mathfrak{sl}_2(\mathbb{C})$ . Hence the Lie algebra arising from the unitary involution on  $M_2(\mathbb{C})$  is of type  $A_1$ .

**Remark 1.2.16.** This explains why, in the  $2 \times 2$  case, the Lie algebra arising from a symplectic involution is isomorphic to the one of type  $A_1$ . Equivalently, in this lowest-rank case one has the accidental isomorphism

$$C_1 \cong A_1.$$

### 1.2.3 They come from the derivation of any algebra

Let  $A$  be any algebra.

**Definition 1.2.17.** A derivation of  $A$  is a linear map  $d : A \rightarrow A$  such that  $d(ab) = d(a)b + ad(b)$  for all  $a, b \in A$ .

**Example 1.2.18.** Let  $A = F[t_1, \dots, t_n]$  be the polynomial algebra. We have the formal derivative  $\partial$  where  $\partial(t_i^n) = nt_i^{n-1}$  and  $\partial(fg) = f\partial(g) + \partial(f)g$ . Let  $d = \sum_{i=1}^n f_i(t_1, \dots, t_n) \frac{\partial}{\partial t_i}$ , this is a derivation of  $A$ .

$$d(f) = \sum_{i=1}^n f_i(t_1, \dots, t_n) \frac{\partial f}{\partial t_i}$$

If  $d_1, d_2$  are derivations of  $A$ , then  $[d_1, d_2]$  is again a derivation of  $A$ . We can check it by the following calculation:

$$\begin{aligned} [d_1, d_2](ab) &= d_1(d_2(ab)) - d_2(d_1(ab)) \\ &= d_1(d_2(a)b + ad_2(b)) - d_2(d_1(a)b + ad_1(b)) \\ &= d_1(d_2(a))b + d_1(ad_2(b)) + d_2(d_1(a))b + d_2(ad_1(b)) \\ &= d_1(d_2(a))b + d_2(d_1(a))b + d_1(ad_2(b)) + d_2(ad_1(b)) \\ &= d_1(d_2(a))b + d_2(d_1(a))b + a[d_1, d_2](b) \\ &= [d_1, d_2](a)b + a[d_1, d_2](b) \\ &= [d_1, d_2](ab) \end{aligned}$$

Therefore  $[d_1, d_2]$  is a derivation of  $A$ .

The above example means that:

$$\text{Der}(A) = \{ d : A \rightarrow A \mid d \text{ is a derivation} \} < \text{End}_F(A)^{(-)}$$

In particular, when  $A = F[t_1, \dots, t_n]$ , we have:

$$\text{Der}(A) = \left\{ \sum_{i=1}^n f_i(t_1, \dots, t_n) \frac{\partial}{\partial t_i} \mid f_i(t_1, \dots, t_n) \in A \right\} < \text{End}_F(A)^{(-)}$$

We denote this algebra by  $W(n)$ . It is called the  $n$ -th Witt algebra. It is a simple, infinite dimensional Lie algebra (when  $\text{char } F = 0$ ).

Thus every element of  $W(n)$  has the form

$$D = \sum_{i=1}^n f_i \partial_i, \quad f_i \in A.$$

**Definition 1.2.19.** *The divergence map is defined by*

$$\operatorname{div} : W(n) \rightarrow A, \quad \operatorname{div} \left( \sum_{i=1}^n f_i \partial_i \right) = \sum_{i=1}^n \partial_i(f_i).$$

*Its kernel is the special Lie algebra*

$$S(n) := \ker(\operatorname{div}) = \left\{ \sum_{i=1}^n f_i \partial_i \in W(n) \mid \sum_{i=1}^n \partial_i(f_i) = 0 \right\}.$$

*Equivalently,  $S(n)$  is the Lie algebra of polynomial vector fields preserving the standard volume form*

$$\omega = dx_1 \wedge \cdots \wedge dx_n.$$

A basic family of elements of  $S(n)$  is given by

$$D_{ij}(f) := \partial_j(f) \partial_i - \partial_i(f) \partial_j, \quad 1 \leq i < j \leq n, \quad f \in A,$$

since

$$\operatorname{div}(D_{ij}(f)) = \partial_i \partial_j(f) - \partial_j \partial_i(f) = 0.$$

One can check

$$S(n) = \langle D_{ij}(f) \mid 1 \leq i < j \leq n, f \in A \rangle_{\operatorname{Lie}}$$

These elements generate  $S(n)$  as a Lie algebra.

**Definition 1.2.20.** *The grading of  $W(n)$  is defined by Give  $A$  its standard grading by declaring  $\deg x_i = 1$ . Then  $W(n)$  becomes a  $\mathbb{Z}$ -graded Lie algebra with*

$$\deg(x^a \partial_i) = |a| - 1, \quad |a| := a_1 + \cdots + a_n,$$

*so that*

$$W(n) = \bigoplus_{k \geq -1} W(n)_k.$$

*and*

$$W(n)_0 = \operatorname{span}_F \{x_j \partial_i \mid 1 \leq i, j \leq n\} \cong \mathfrak{gl}_n(F).$$

Since the divergence map is homogeneous of degree 0, the subalgebra  $S(n)$  is graded:

$$S(n) = \bigoplus_{k \geq -1} S(n)_k, \quad S(n)_k = S(n) \cap W(n)_k.$$

Moreover,

$$S(n)_{-1} = W(n)_{-1},$$

and

$$S(n)_0 = \left\{ \sum_{i,j} a_{ij} x_j \partial_i \mid \sum_i a_{ii} = 0 \right\} \cong \mathfrak{sl}_n(F),$$

because

$$\operatorname{div} \left( \sum_{i,j} a_{ij} x_j \partial_i \right) = \sum_i a_{ii}.$$

In characteristic 0, the special algebra is perfect:

$$[S(n), S(n)] = S(n).$$

Hence

$$S(n)^{(1)} = S(n).$$

For  $n \geq 2$ , the Lie algebra  $S(n)$  is simple. Therefore, in characteristic 0 one has

$$W(n) \supset S(n) = S(n)^{(1)},$$

where both algebras are infinite-dimensional,  $W(n)$  is the full Lie algebra of polynomial vector fields, and  $S(n)$  is its simple divergence-zero subalgebra.

**Remark 1.2.21.** *In characteristic 0, the chain  $W(n) \supset S(n) \supset S(n)^{(1)}$  is thus less informative than in positive characteristic, since the last inclusion is an equality.*

Let  $A$  be an associative algebra. Fix an element  $a \in A$ , we define  $\operatorname{ad}(a) : A \rightarrow A : b \mapsto [a, b]$ . This is a derivation of  $A$ .

**Definition 1.2.22.** *Let  $L$  be a Lie algebra. The adjoint map is defined as:*

$$\operatorname{ad} : L \rightarrow \operatorname{End}_F(L)^{(-)}, a \mapsto \operatorname{ad}(a)$$

*where  $\operatorname{ad}(a) : L \rightarrow L : b \mapsto [a, b]$  is an inner derivation of  $L$ . We also call  $\operatorname{ad}(a)$  an adjoint operator.*

Note that  $\operatorname{ad}(a) \in \operatorname{Der}(L)$  check it by the following calculation:

$$\begin{aligned} \operatorname{ad}(a)([b, c]) &= \operatorname{ad}(a)(bc - cb) \\ &= [a, bc] - [a, cb] \\ &= [a, b]c + b[a, c] - [a, c]b - c[a, b] \\ &= [\operatorname{ad}(a)(b), c] + [b, \operatorname{ad}(a)(c)] \end{aligned}$$

”Inner” here comes from that  $\operatorname{ad}(L) \triangleleft \operatorname{Der}(L)$  since  $\forall d \in \operatorname{Der}(L)$ , we have  $[d, \operatorname{ad}(a)] =$

$\text{ad}(d(a))$ . Check it by the following calculation:

$$\begin{aligned} [d, \text{ad}(a)](x) &= d([a, x]) - [a, d(x)] \\ &= [d(a), x] + [a, d(x)] - [a, d(x)] \\ &= [d(a), x] \\ &= \text{ad}(d(a))(x). \end{aligned}$$

**Remark 1.2.23** (Exponentials of nilpotent inner derivations). *Let  $L$  be a Lie algebra over a field  $F$ , and let*

$$d \in \text{ad}(L) \subseteq \text{Der}(L).$$

*Thus there exists  $a \in L$  such that*

$$d = \text{ad}(a), \quad d(x) = [a, x] \quad \text{for all } x \in L.$$

*Assume that  $d$  is nilpotent, so that  $d^N = 0$  for some integer  $N \geq 1$ . Then one defines the exponential of  $d$  by*

$$\exp(d) := \sum_{n=0}^{\infty} \frac{d^n}{n!}.$$

*Since  $d$  is nilpotent, this is actually a finite sum:*

$$\exp(d) = I + d + \frac{d^2}{2!} + \cdots + \frac{d^{N-1}}{(N-1)!}.$$

*Hence  $\exp(d)$  is a well-defined linear endomorphism of  $L$ .*

*Moreover,  $\exp(d)$  is invertible, with inverse*

$$\exp(d)^{-1} = \exp(-d).$$

*Since  $d$  is a derivation,  $\exp(d)$  is in fact a Lie algebra automorphism:*

$$\exp(d)([x, y]) = [\exp(d)(x), \exp(d)(y)] \quad \text{for all } x, y \in L.$$

*Therefore*

$$\exp(d) \in \text{Aut}(L).$$

*In particular, if  $d = \text{ad}(a)$ , then*

$$\exp(\text{ad}(a))(x) = x + [a, x] + \frac{1}{2!}[a, [a, x]] + \frac{1}{3!}[a, [a, [a, x]]] + \cdots.$$

*When  $\text{ad}(a)$  is nilpotent, this sum is finite. The automorphism*

$$\exp(\text{ad}(a))$$

*is called an inner automorphism arising from the nilpotent inner derivation  $\text{ad}(a)$ .*

A fundamental example is the matrix Lie algebra  $L = \mathfrak{gl}_n(F)$ . For  $A \in \mathfrak{gl}_n(F)$ , one has

$$\mathrm{ad}(A)(X) = [A, X] = AX - XA,$$

and

$$\exp(\mathrm{ad}(A))(X) = e^A X e^{-A}.$$

Thus, in the matrix case, the exponential of an inner derivation is exactly conjugation by the matrix exponential.

**Proposition 1.2.24.** *Let  $L$  be a Lie algebra over a field  $F$ , and let  $d \in \mathrm{Der}(L)$  be nilpotent. Then*

$$\exp(d) := \sum_{n=0}^{\infty} \frac{d^n}{n!}$$

*is a Lie algebra automorphism of  $L$ .*

*Proof.* Since  $d$  is nilpotent, there exists  $N \geq 1$  such that  $d^N = 0$ . Hence

$$\exp(d) = I + d + \frac{d^2}{2!} + \cdots + \frac{d^{N-1}}{(N-1)!}$$

is a finite sum, so it is a well-defined linear endomorphism of  $L$ .

Moreover,

$$\exp(d) \exp(-d) = \exp(-d) \exp(d) = I,$$

because  $d$  commutes with itself and the usual exponential identity holds for polynomials in  $d$ . Thus  $\exp(d)$  is invertible, with inverse  $\exp(-d)$ .

It remains to show that  $\exp(d)$  preserves the Lie bracket. Define

$$\phi_t := \exp(td) = \sum_{n=0}^{\infty} \frac{t^n d^n}{n!},$$

where the sum is again finite since  $d$  is nilpotent. For  $x, y \in L$ , set

$$F(t) := \phi_t([x, y]), \quad G(t) := [\phi_t(x), \phi_t(y)].$$

We show that  $F(t) = G(t)$ .

First,

$$F'(t) = d(\phi_t([x, y])) = \phi_t(d([x, y])).$$

Since  $d$  is a derivation,

$$d([x, y]) = [d(x), y] + [x, d(y)].$$

Therefore

$$F'(t) = \phi_t([d(x), y] + [x, d(y)]) = [\phi_t(d(x)), \phi_t(y)] + [\phi_t(x), \phi_t(d(y))].$$

On the other hand,

$$G'(t) = [\phi'_t(x), \phi_t(y)] + [\phi_t(x), \phi'_t(y)] = [\phi_t(d(x)), \phi_t(y)] + [\phi_t(x), \phi_t(d(y))].$$

Hence

$$F'(t) = G'(t).$$

Also,

$$F(0) = [x, y] = G(0).$$

It follows that  $F(t) = G(t)$  for all  $t$ , and in particular at  $t = 1$  we get

$$\exp(d)([x, y]) = [\exp(d)(x), \exp(d)(y)].$$

Thus  $\exp(d)$  is a Lie algebra automorphism of  $L$ . □

#### 1.2.4 They come from brackets

**Definition 1.2.25** (Poisson bracket). *Let  $A$  be a commutative associative algebra over a field  $\mathbb{k}$ . A Poisson bracket on  $A$  is a bilinear map*

$$\{\cdot, \cdot\} : A \times A \rightarrow A$$

*such that for all  $f, g, h \in A$  the following conditions hold:*

1. **Skew-symmetry:**

$$\{f, g\} = -\{g, f\}.$$

2. **Jacobi identity:**

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0.$$

3. **Leibniz rule:**

$$\{f, gh\} = \{f, g\}h + g\{f, h\}.$$

*In other words,  $(A, \{\cdot, \cdot\})$  is a Lie algebra, and for each fixed  $f \in A$ , the map*

$$\text{ad}_f : A \rightarrow A, \quad \text{ad}_f(g) = \{f, g\},$$

*is a derivation of the algebra  $A$ . An algebra  $A$  equipped with such a bracket is called a Poisson algebra.*

**Remark 1.2.26.** *The Leibniz rule in the second variable follows automatically from skew-symmetry:*

$$\{fg, h\} = f\{g, h\} + g\{f, h\}.$$

*Thus a Poisson bracket is precisely a Lie bracket compatible with the commutative multiplication on  $A$  by the derivation property.*

**Example 1.2.27** (Canonical Poisson bracket on  $\mathbb{R}^{2n}$ ). Let

$$A = C^\infty(\mathbb{R}^{2n}), \quad (q_1, \dots, q_n, p_1, \dots, p_n)$$

be the standard coordinates. Define

$$\{f, g\} = \sum_{i=1}^n \left( \frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right).$$

Then  $\{\cdot, \cdot\}$  is a Poisson bracket on  $A$ . In particular,

$$\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}.$$

**Definition 1.2.28.** Let  $F$  be a field of characteristic 0, and let  $\mathcal{A}$  be an algebra of differentiable functions in one variable  $x$ . For  $f, g \in \mathcal{A}$ , define

$$[f, g] := fg' - f'g,$$

where  $f' = \frac{df}{dx}$  and  $g' = \frac{dg}{dx}$ .

**Definition 1.2.29.** To each  $f \in \mathcal{A}$ , associate the vector field

$$X_f := f(x) \frac{d}{dx}.$$

**Proposition 1.2.30.** For  $f, g \in \mathcal{A}$ , one has

$$[X_f, X_g] = X_{[f, g]}.$$

Equivalently,

$$[X_f, X_g] = (fg' - gf') \frac{d}{dx}.$$

*Proof.* For any test function  $u$ , we have

$$X_f(u) = fu', \quad X_g(u) = gu'.$$

Thus

$$X_f(X_g(u)) = f(gu')' = fg'u' + fg u'',$$

and similarly

$$X_g(X_f(u)) = g(fu')' = gf'u' + gf u''.$$

Subtracting, we obtain

$$[X_f, X_g](u) = X_f(X_g(u)) - X_g(X_f(u)) = (fg' - gf')u'.$$

Therefore

$$[X_f, X_g] = (fg' - gf') \frac{d}{dx} = X_{[f,g]}.$$

□

**Proposition 1.2.31.** *The operation*

$$[f, g] = fg' - f'g$$

*is bilinear and skew-symmetric.*

*Proof.* Bilinearity follows immediately from the linearity of differentiation. Also,

$$[g, f] = gf' - g'f = -(fg' - f'g) = -[f, g].$$

Hence the bracket is skew-symmetric. □

**Theorem 1.2.32.** *The bracket*

$$[f, g] = fg' - f'g$$

*defines a Lie algebra structure on  $\mathcal{A}$ .*

*Proof.* By the previous proposition, it remains only to verify the Jacobi identity. This follows from the fact that the map

$$f \mapsto X_f = f \frac{d}{dx}$$

identifies this bracket with the usual Lie bracket of vector fields on the line, and the latter always satisfies the Jacobi identity. Hence

$$[f, [g, h]] + [g, [h, f]] + [h, [f, g]] = 0$$

for all  $f, g, h \in \mathcal{A}$ . □

### 1.2.5 They come from Lie groups

**Definition 1.2.33.** *Let  $G$  be a Lie group with identity element  $e$ . The Lie algebra of  $G$ , denoted by  $\text{Lie}(G)$  or  $\mathfrak{g}$ , is the tangent space*

$$\mathfrak{g} := T_e G$$

*equipped with the bracket induced by left-invariant vector fields.*

**Definition 1.2.34.** For each  $X \in T_e G$ , the corresponding left-invariant vector field  $\tilde{X}$  on  $G$  is defined by

$$\tilde{X}_g := (dL_g)_e(X), \quad g \in G,$$

where  $L_g : G \rightarrow G$  is left translation,  $L_g(h) = gh$ .

**Proposition 1.2.35.** The assignment  $X \mapsto \tilde{X}$  gives a bijection between  $T_e G$  and the space of left-invariant vector fields on  $G$ .

**Definition 1.2.36.** For  $X, Y \in \mathfrak{g} = T_e G$ , define

$$[X, Y] := [\tilde{X}, \tilde{Y}](e),$$

where on the right-hand side  $[\tilde{X}, \tilde{Y}]$  is the usual Lie bracket of vector fields on  $G$ .

**Proposition 1.2.37.** The bracket defined above makes  $T_e G$  into a Lie algebra. In particular, it is bilinear, alternating, and satisfies the Jacobi identity.

**Example 1.2.38.** If  $G = GL_n(F)$ , where  $F = \mathbb{R}$  or  $\mathbb{C}$ , then

$$\text{Lie}(G) = T_I GL_n(F) \cong M_n(F),$$

and the Lie bracket is

$$[X, Y] = XY - YX.$$

Hence

$$\text{Lie}(GL_n(F)) = \mathfrak{gl}_n(F).$$

**Example 1.2.39.** If  $G = SL_n(F)$ , then

$$\text{Lie}(SL_n(F)) = \mathfrak{sl}_n(F) = \{X \in M_n(F) \mid \text{tr}(X) = 0\}.$$

Indeed, differentiating the defining equation  $\det(g) = 1$  at the identity gives the trace-zero condition.

**Example 1.2.40.** If  $G = O_n(\mathbb{R})$ , then

$$\text{Lie}(O_n(\mathbb{R})) = \mathfrak{o}_n(\mathbb{R}) = \{X \in M_n(\mathbb{R}) \mid X^T = -X\}.$$

This follows by differentiating the relation  $g^T g = I$  at the identity.

### 1.3 Basic concepts of Lie algebra

Let  $L$  be a Lie algebra over a field  $F$ .

**Definition 1.3.1.** The derived series of  $L$  is defined inductively by

$$L^{(0)} = L, \quad L^{(1)} = [L, L], \quad L^{(i)} = [L^{(i-1)}, L^{(i-1)}] \quad (i \geq 1).$$

The Lie algebra  $L$  is called solvable if

$$L^{(n)} = 0$$

for some integer  $n \geq 0$ .

**Definition 1.3.2.** The descending central series of  $L$  is defined by

$$L^1 = L, \quad L^2 = [L, L], \quad L^i = [L, L^{i-1}] \quad (i \geq 2).$$

The Lie algebra  $L$  is called nilpotent if

$$L^n = 0$$

for some integer  $n \geq 1$ .

**Proposition 1.3.3.** If  $L$  is nilpotent, then  $L$  is solvable.

*Proof.* It is enough to show that

$$L^{(i)} \subseteq L^{2^i} \quad (i \geq 0).$$

We argue by induction on  $i$ . The case  $i = 0$  is clear. If

$$L^{(i)} \subseteq L^{2^i},$$

then

$$L^{(i+1)} = [L^{(i)}, L^{(i)}] \subseteq [L^{2^i}, L^{2^i}] \subseteq L^{2^{i+1}}.$$

Hence the claim follows for all  $i$ . If  $L^n = 0$  for some  $n$ , then for  $i$  sufficiently large one has  $2^i \geq n$ , so

$$L^{(i)} \subseteq L^{2^i} = 0.$$

Therefore  $L$  is solvable. □

**Definition 1.3.4.** A solvable ideal of  $L$  is an ideal which is solvable as a Lie algebra in its own right. Since the sum of two solvable ideals is again solvable, there exists a

unique maximal solvable ideal of  $L$ , called the radical of  $L$ , denoted by

$$\text{Rad}(L).$$

The Lie algebra  $L$  is called semisimple if

$$\text{Rad}(L) = 0.$$

**Remark 1.3.5.** Thus the three notions fit into the chain

$$\text{nilpotent} \implies \text{solvable},$$

while semisimplicity means that  $L$  has no nonzero solvable ideals. In particular, a semisimple Lie algebra cannot contain any nonzero abelian ideal.

**Example 1.3.6.** Any abelian Lie algebra is nilpotent, hence solvable. On the other hand, the Lie algebra of upper triangular matrices is solvable in general, but need not be nilpotent.

**Example 1.3.7.** Let

$$L = \left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \mid a, b \in F \right\} \subseteq \mathfrak{gl}_2(F),$$

Then  $L$  is solvable but not nilpotent.

Indeed, if

$$x = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix},$$

then

$$[x, y] = \begin{pmatrix} 0 & ad - bc \\ 0 & 0 \end{pmatrix}.$$

Hence

$$[L, L] = \left\{ \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \mid t \in F \right\}, \quad [[L, L], [L, L]] = 0.$$

Thus the derived series terminates:

$$L^{(1)} = [L, L], \quad L^{(2)} = [L^{(1)}, L^{(1)}] = 0,$$

so  $L$  is solvable.

On the other hand,  $L$  is not nilpotent, since its descending central series does not reach 0. In fact,

$$L^2 = [L, L] = \left\{ \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \mid t \in F \right\},$$

and for any

$$x = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \in L, \quad y = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \in L^2,$$

we have

$$[x, y] = \begin{pmatrix} 0 & at \\ 0 & 0 \end{pmatrix} \in L^2.$$

Therefore

$$L^3 = [L, L^2] = L^2,$$

and inductively

$$L^n = L^2 \neq 0 \quad (n \geq 2).$$

So  $L$  is not nilpotent.

## Chapter 2

# Lecture 2 Engel Theorem

### 2.1 Engel Theorem the first type

**Definition 2.1.1.** Let  $L$  be a Lie algebra over a field  $F$ . An enveloping algebra of  $L$  is an associative  $F$ -algebra  $A$  together with a Lie algebra homomorphism

$$\iota : L \rightarrow A^-,$$

where  $A^-$  denotes the Lie algebra obtained from  $A$  by equipping the underlying vector space of  $A$  with the bracket

$$[x, y] = xy - yx,$$

such that the associative algebra  $A$  is generated by  $\iota(L)$  (or just writing  $A = \langle L \rangle_{\text{assoc}}$  or simply  $A = \langle L \rangle$ ).

If, in addition, the pair  $(A, \iota)$  satisfies the following universal property: for every associative  $F$ -algebra  $B$  and every Lie algebra homomorphism

$$\varphi : L \rightarrow B^-,$$

there exists a unique associative algebra homomorphism

$$\tilde{\varphi} : A \rightarrow B$$

such that

$$\varphi = \tilde{\varphi} \circ \iota,$$

then  $A$  is called the universal enveloping algebra of  $L$ .

**Theorem 2.1.2.** Let  $A$  be a finite dimensional enveloping algebra of  $L$ , i.e.  $L \subseteq A^{(-)}$ ,  $A = \langle L \rangle$ . If  $\exists n \geq 1$  such that  $\forall a \in L, a^n = 0$ . Then  $A$  is nilpotent.

## 2.2 Engel Theorem the second type

In fact we can prove a stronger version.

**Definition 2.2.1.** A subset  $S \subseteq L$  is called a Lie subset if

$$[a, b] \in S \quad \text{for all } a, b \in S.$$

Note that a Lie subset need not be a subspace, and hence need not be a Lie subalgebra.

**Theorem 2.2.2.** Let  $A$  be a finite-dimensional associative algebra over  $F$ , and let

$$L \subseteq A^{(-)}$$

be a Lie subalgebra such that  $A$  is generated by  $L$  as an associative algebra. Suppose that

$$L = \text{span}(S)$$

for some Lie subset  $S \subseteq L$ , and that every element of  $S$  is nilpotent in  $A$ . Then  $A$  is a nilpotent associative algebra.

**Remark 2.2.3.** In the finite-dimensional situation, the existence of a uniform exponent  $n$  is automatic once every element of  $S$  is nilpotent in  $A$ . Indeed, let

$$m = \dim_F A,$$

and let  $a \in A$  be nilpotent. Consider the left multiplication operator

$$L_a : A \rightarrow A, \quad x \mapsto ax.$$

Since  $a$  is nilpotent,  $L_a$  is a nilpotent linear transformation on the  $m$ -dimensional vector space  $A$ . Hence

$$L_a^m = 0.$$

But

$$L_a^m = L_{a^m},$$

so  $a^m = 0$ . Therefore every nilpotent element of  $A$  has nilpotency index at most  $\dim_F A$ . In particular, if every element of  $S$  is nilpotent in  $A$ , then one may take

$$n = \dim_F A$$

and obtain

$$a^n = 0 \quad \text{for all } a \in S.$$

Thus, in the theorem above, the hypothesis that every element of  $S$  is nilpotent in  $A$  already implies the existence of a common exponent  $n$ .

**Theorem 2.2.4** (Zorn's Lemma). *Let  $(P, \leq)$  be a partially ordered set. Assume that every chain in  $P$  has an upper bound in  $P$ . Then  $P$  contains a maximal element.*

*Proof.* We prove the result assuming the Well-Ordering Theorem.

Suppose, for contradiction, that  $P$  has no maximal element. Then for every  $x \in P$  there exists some  $y \in P$  such that

$$x < y.$$

By the Well-Ordering Theorem, we may choose a well-ordering  $\preceq$  on  $P$ . For each  $x \in P$ , let  $f(x)$  be the  $\preceq$ -least element of the set

$$\{y \in P : x < y\}.$$

Then

$$x < f(x) \quad \text{for all } x \in P.$$

We now define, by transfinite recursion, a family  $(x_\alpha)$  indexed by ordinals. Suppose that for all  $\beta < \alpha$  the elements  $x_\beta$  have been defined and form a chain. By hypothesis, the chain

$$C_\alpha := \{x_\beta : \beta < \alpha\}$$

has an upper bound  $u_\alpha \in P$ . Define

$$x_\alpha := f(u_\alpha).$$

Since  $u_\alpha$  is an upper bound of  $C_\alpha$ , for every  $\beta < \alpha$  we have

$$x_\beta \leq u_\alpha < x_\alpha.$$

Hence  $(x_\alpha)$  is a strictly increasing family:

$$\beta < \alpha \implies x_\beta < x_\alpha.$$

Therefore the map

$$\alpha \longmapsto x_\alpha$$

is injective from the class of all ordinals into the set  $P$ . This is impossible, since there is no injection from the class of all ordinals into a set.

This contradiction shows that  $P$  must contain a maximal element.  $\square$

**Lemma 2.2.5.** *Let*

$$\mathcal{P} = \{T \subseteq L \mid T \text{ is a Lie subset and } a^n = 0 \text{ for all } a \in T\},$$

*partially ordered by inclusion. Then  $\mathcal{P}$  has a maximal element.*

*Proof.* We apply Zorn's lemma. Let  $\{T_i\}_{i \in I}$  be a chain in  $\mathcal{P}$ , and set

$$T = \bigcup_{i \in I} T_i.$$

We claim that  $T \in \mathcal{P}$ .

First,  $T$  is a Lie subset. Indeed, if  $a, b \in T$ , then  $a \in T_i$  and  $b \in T_j$  for some  $i, j \in I$ . Since  $\{T_i\}_{i \in I}$  is a chain, one of  $T_i, T_j$  is contained in the other; say  $T_i \subseteq T_j$ . Then  $a, b \in T_j$ , and since  $T_j$  is a Lie subset,

$$[a, b] \in T_j \subseteq T.$$

Second, if  $a \in T$ , then  $a \in T_i$  for some  $i \in I$ , so  $a^n = 0$  because  $T_i \in \mathcal{P}$ .

Thus  $T \in \mathcal{P}$ , and clearly  $T$  is an upper bound of the chain. Therefore every chain in  $\mathcal{P}$  has an upper bound. By Zorn's lemma,  $\mathcal{P}$  has a maximal element.  $\square$